

Draft: the finite excess in transfinite nim multiplication

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June 2026

Abstract

Conway's field On_2 makes the ordinals below ω^{ω^ω} an algebraic closure of $\mathbb{F}_2$ under nim-arithmetic. Its multiplication is governed by a Kummer tower whose carries are Lenstra's *excess* values: for each odd prime p , the carry is $\alpha_p = \kappa_{f(p)} + m_p$, where the transfinite part $\kappa_{f(p)}$ has a known closed shape (Lenstra, DiMuro) but the finite correction m_p does not. This note consolidates the state of the excess problem (docs/OPEN.md Problem 3 of ogdoad): the exact finite-field reformulation of the root test as a multiplicative-order criterion and its structural norm reduction; a complete analysis of the 3-power column via a half-angle splitting in the cyclotomic component fields, certifying $m_r = 1$ for all primes r with $f(r) \in \{3, 9, \dots, 3^6\}$ and forcing $m_p \geq 4$ unconditionally on the $f(p) = 2 \cdot 3^k$ column — now sharpened to $m_p = 4$ *exactly* at every prime current factor tables reach (universally for $k \leq 6$), via a corrected compositum norm $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega$ that collapses the $m = 4$ test into the same trinomial field; the empirically exact 0/1/4 candidate rule and the proof that the Lenstra component set alone cannot decide the excess; and the reduction of Lenstra's open boundedness question to that candidate. Feasibility of the first open row, $p = 719$, is assessed: a Frobenius-orbit norm recurrence reduces it to arithmetic in $\mathbb{F}_{2^{359}}$, with the cost concentrated in tower-aware Frobenius composition. Each statement carries an explicit claim-level tag.

Claim levels. [**proved**] – unconditional, machine-checked or classically verified; [**certified** $k \leq 6$] – proved for the stated range, blocked beyond it only by unfactored cofactors; [**consistent**] – no counterexample found, not proved; [**conjectured**] – explicitly open in the literature or posed in this thread; [**open**] – neither confirmed nor refuted, no feasible attack on record.

1 Notation, implemented state, external data

1.1 Notation

For an odd prime p :

- $f(p) = \text{ord}_p(2)$, the multiplicative order of 2 modulo p ;
- κ_h is the Lenstra/DiMuro tower element indexed by h ;
- $\mathcal{Q}(h)$ is Lenstra's set of prime-power components appearing in κ_h ;
- the *excess* m_p is the least finite m such that $\kappa_{f(p)} + m$ has no p -th root in the relevant finite component field;
- the Kummer carry is $\alpha_p = \kappa_{f(p)} + m_p$.

For the ordinal tower, a row with $\mathcal{Q}(f(p)) = \{q\}$ and finite excess m gives the ordinal sum corresponding to $\kappa_q + m$. Addition of nimbers is XOR; $\otimes$ is the nim product.

1.2 Implemented tower state

In the Rust tower (`src/scalar/big/ordinal/tower.rs`), as of 2026-06-13:

- the finite excess rows m_p are now sourced from OEIS A380496 [7], the b-file’s 126 known rows (odd primes $3 \leq p \leq 709$); the first 14 rows reproduce DiMuro Table 1 + the formerly locally verified $\alpha_{47} = \omega^{(\omega^7)} + 1$;
- the operational boundary is α_{719} , the first OEIS-unknown row: a carry needing α_{719} or beyond returns **None**;
- the table extends *reach*, not *feasibility*: each α_p is reconstructed in code from $\text{ord}_p(2)$, $\mathcal{Q}(f(p))$, and m_p , but for large p the $\mathcal{Q}$ /finite-subfield reconstruction lives in the degree- e_p component field (e_p in the millions; see the snapshot below), so only the small- e_p rows are materializable end-to-end today.

1.3 External data snapshot (2026-06-09)

- OEIS A380496 [7] has 1417 extended rows: 799 known and 618 unknown; the b-file has 126 initial known rows.
- The first OEIS unknown is row $n = 127$, the 127th odd prime $p = 719$. For $p = 719$: $f(719) = \text{ord}_{719}(2) = 359$ and $\mathcal{Q}(359) = \{359\}$.
- The transfinite-nim-calculator logs [9] record the direct component exponent as $e_{719} = 1,258,230,380$, the practical wall for direct exponentiation.

2 The order criterion and structural norm reduction

2.1 The exact criterion

Throughout, E denotes the degree of the component field containing $\beta = \kappa_{f(p)} + m$, so $f = f(p) \mid E$ and hence $p \mid 2^E - 1$.

Proposition 1 (power criterion). *[proved] $\beta \in \mathbb{F}_{2^E}^\times$ has no p -th root in $\mathbb{F}_{2^E}$ iff $\beta^{(2^E-1)/p} \neq 1$. Hence*

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

Proof. $\mathbb{F}_{2^E}^\times$ is cyclic of order $2^E - 1$ and $p \mid 2^E - 1$; in a cyclic group of order N , β is a p -th power iff $\beta^{N/p} = 1$. $\square$

Corollary 1 (order form). *[proved] If $v_p(2^E - 1) = 1$, then β has no p -th root in $\mathbb{F}_{2^E}$ iff $p \mid \text{ord}(\beta)$, so m_p is the least m with $p \mid \text{ord}(\kappa_{f(p)} + m)$.*

Remark 1 (Wieferich caveat). In general “no p -th root” is equivalent to $v_p(\text{ord } \beta) = v_p(2^E - 1)$, which collapses to $p \mid \text{ord}(\beta)$ only when $v_p(2^E - 1) = 1$. Since $v_p(2^E - 1) = v_p(2^{f(p)} - 1)$ whenever $p \nmid E/f(p)$ (lifting the exponent), the hypothesis can fail only at base-2 Wieferich-type primes with $p^2 \mid 2^{f(p)} - 1$; the only known instances are $p = 1093$ and $p = 3511$, both of which lie inside the extended A380496 range. For those rows the exact test of Proposition 1 (or the full- p -part order condition) must be used. The certifications of Section 3 compute orders exactly, with squarefreeness checks, and are unaffected. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

2.2 Norm reduction

Proposition 2 (norm reduction). *[proved] For $\beta \in \mathbb{F}_{2^E}^\times$ with $f = f(p) \mid E$,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

Proof. $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$, and $((2^E-1)/(2^f-1)) \cdot ((2^f-1)/p) = (2^E-1)/p$. $\square$

So the $p = 719$ test reduces to: (1) compute $N = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^{359}}}(\kappa_{359} + 1)$ in $\mathbb{F}_{2^{359}}$; (2) test whether $719 \mid \text{ord}(N)$ in $\mathbb{F}_{2^{359}}^\times$, a field of degree 359 in which $719 \mid 2^{359} - 1$.

Proposition 3 (Galois-conjugacy invariance). *[proved] The order of $\kappa_{f(p)} + m$ is constant on its Frobenius orbit in $\mathbb{F}_{2^E}$, and the norm $N = \prod_{i=0}^{E/f-1} \text{Frob}^{if}(\kappa_f + m)$ is determined structurally by the tower equations without materialising the full orbit.*

Proposition 4 (tower pinning). *[proved] The tower equations pin κ_{359} up to Frobenius conjugacy: each composed minimal polynomial along the component chain $359 \rightarrow 179 \rightarrow 89 \rightarrow 11 \rightarrow 5 \rightarrow \text{finite}$ is irreducible (the non- q -th-power property defining the lower excess guarantees this), so κ_{359} is a well-defined conjugacy class and N is a well-defined element of $\mathbb{F}_{2^{359}}$.*

2.3 Feasibility for $p = 719$

The component field degree is $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359 = 1,258,230,380$; the norm requires $E/f = 3,504,820$ Frobenius steps.

[consistent] The norm can in principle be computed by the binary-splitting Frobenius recurrence $N_{2k} = N_k \cdot \text{Frob}^{f^k}(N_k)$ (iterated-Frobenius technique, von zur Gathen–Shoup [10], Kaltofen–Shoup [4]), requiring about $\log_2(3,504,820) \approx 22$ squarings-with-Frobenius-composition at degree E over $\mathbb{F}_2$. Each Frobenius composition is a modular composition in $\mathbb{F}_2[x]$ modulo a degree- E polynomial; with dense polynomial arithmetic at $E \approx 1.26 \times 10^9$ this is not locally feasible.

[conjectured] (as a cost model, not a theorem) Kummer-tower-aware Frobenius – standard-lattice arithmetic for compatibly embedded finite fields in the style of De Feo–Randriam–Rousseau [2], where q -power Frobenius acts near-monomially per tower level – is the likely 10–100 $\times$ lever: the sparse factored structure $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359$ admits a stepwise Frobenius representation in which each level contributes a small-degree modular composition. A *measured* cost model, not a run, is the correct first deliverable.

[open] Whether the Kummer-tower Frobenius model makes the m_{719} computation feasible in wall-clock hours rather than months on available hardware (a single consumer GPU; GF(2) arithmetic is XOR-sliced and GPU-friendly).

[proved] (2026-06-14 local rehearsal.) The first two dependency rows for the $p = 719$ norm path are locally certified by `experiments/ordinal_excess_probe.py`: $m_{89} = 1$ in the $E = 220$ component field, and $m_{179} = 1$ in the $E = 19,580$ component field via the fixed-base power path (`--deep`, about one minute locally). The remaining local rehearsal row is $m_{359} = 1$, with $E = 3,504,820$. This row is already source-pinned by the imported A380496 table ($359 \leq 709$); what remains open here is an independent certificate on the same code path needed for the genuinely unknown $p = 719$ row.

2.4 The m_{359} rehearsal obstruction

Set

$$d = 19,580, \quad E = 179d = 3,504,820, \quad B = \mathbb{F}_{2^d}, \quad F = \mathbb{F}_{2^E}, \quad L = \mathbb{F}_{2^{179}}.$$

For the $m_{359} = 1$ rehearsal, $f(359) = 179$, $\mathcal{Q}(f(359)) = \mathcal{Q}(179) = \{179\}$, and the candidate element is $\beta = \kappa_{179} + 1$. To represent κ_{179} , the tower uses the lower prime row $f(179) = 178$, $\mathcal{Q}(178) = \{89\}$, $m_{179} = 1$. Thus in the tower model $F = B(x)$ with $x = \kappa_{179}$ and

$$x^{179} = \kappa_{89} + 1 \in B,$$

because the lower row $m_{179} = 1$ has already been certified.

[proved] The cheap Kummer norm down the top tower step is not the 359-test norm. Indeed,

$$N_{F/B}(x + 1) = 1 + (\kappa_{89} + 1) = \kappa_{89}$$

(evaluate the monic polynomial $T^{179} + (\kappa_{89} + 1)$ at $T = 1$, in characteristic 2). But $359 \nmid |B^\times| = 2^{19580} - 1$, since $\text{ord}_{359}(2) = 179$ and $179 \nmid 19580$. Thus the structurally cheap norm lands in exactly the field where the 359-primary obstruction is invisible.

[proved] The norm forced by Proposition 2 is the transverse one

$$N_{F/L}(\beta) = \prod_{i=0}^{19579} \text{Frob}^{179i}(\beta) \in L.$$

Here $\gcd(179, 19580) = 1$, so $B \cap L = \mathbb{F}_2$ and $F = BL$. The Frobenius Frob^{179} preserves B as a set but acts nontrivially on it; it is not the relative Kummer action $x \mapsto \zeta x$ over B . Consequently the known lower certificates for m_{89} and m_{179} do not propagate to m_{359} by taking the easy top-step norm.

[proved] The Wieferich caveat is absent at the next two pressure points: direct modular checks give $2^{179} \not\equiv 1 \pmod{359^2}$ and $2^{359} \not\equiv 1 \pmod{719^2}$. The order form of Corollary 1 is therefore equivalent to the full power criterion for the m_{359} rehearsal and the proposed m_{719} test.

[consistent] (2026-06-16 diagnostic.) Re-running the maintained probe confirmed the quick certificate table and the `--deep` $m_{179} = 1$ certificate. A side norm experiment then computed the correct transverse norm in the current term basis for the feasible analogues: for $p = 89$ the norm to $\mathbb{F}_{2^{11}}$ has support 111/220, and for $p = 179$ the norm to $\mathbb{F}_{2^{178}}$ has support 9691/19580. Thus the target-subfield element is essentially half-dense in this representation; using it as the new fixed base makes the final root-test exponent slower than the direct fixed-base certificate. This is a representation-level diagnostic, not a theorem that no sparse description exists.

[open] The current pure-Python term algebra has no feasible route to a local m_{359} certificate. The direct fixed-base path would have to precompute multiplication by β for $E = 3,504,820$ basis terms and then run an E -bit exponent. The norm path lowers the mathematical problem to a degree-179 target field, but computing that norm still requires the 19580-term transverse Frobenius orbit through $F_{2^{3504820}}$. A practical certificate now appears to require either (i) dense/sliced GF(2) arithmetic such as `gf2x/NTL`, or (ii) a tower-aware Frobenius representation that makes the transverse orbit cheap. This is the obstruction hit in the present pass.

Remark 2 (shared abstraction). The structural primitive needed here – $N_{E/K}(\beta) = \prod_i \text{Frob}^i(\beta)$ computed by a Frobenius-orbit recurrence – is the same primitive that the cyclic-algebra Brauer-invariant bridge (Bridge K, `docs/TBD.md`) requires for reduced norms over a Frobenius-generated cyclic extension. The existing `FieldExtension::relative_norm` does not apply to the term-algebra/transfinite setting, but the shape is identical; factoring out a reusable `relative_norm_over_frobenius_orbit` is an engineering opportunity. This is not a claim that the bounded `Fpn` norm certifies m_{719} .

3 The 3^k family

Write $\zeta := \kappa_{3^k}$ and $h := 3^k$ throughout this section.

3.1 Structure

Proposition 5 (cyclotomic recognition). *[proved] $\zeta^{3^k} = \kappa_2$, which has order 3; hence ζ is a primitive 3^{k+1} -st root of unity. Since 2 is a primitive root modulo 3^{k+1} , the cyclotomic polynomial $\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$ is irreducible over $\mathbb{F}_2$, the component field is $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$, and it carries the index-2 subfield $L = \mathbb{F}_{2^h}$.*

Proposition 6 (half-angle splitting). *[proved] With $s = (3^{k+1} + 1)/2$ (the inverse of 2 modulo 3^{k+1}),*

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

where ζ^s lies in the norm-one circle U (of order exactly 3^{k+1} within the subgroup of order $2^h + 1$) and $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders,

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

Machine-checked corollaries: $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$ and $N_{F/L}(\kappa + 1) = \gamma_k$.

Proposition 7 (translates $m = 2, 3$). *[proved] $\kappa_{3^k} + 2$ and $\kappa_{3^k} + 3$ split as in Proposition 6 with γ_k replaced by a Galois conjugate. Therefore $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$ for $m = 1, 2, 3$, and $\text{ord}(\kappa_{3^k}) = 3^{k+1}$.*

Proposition 8 (the $2 \cdot 3^k$ exception, unconditionally). *[proved] Any prime p with $f(p) = 2 \cdot 3^k$ satisfies $p \mid 2^{3^k} + 1$, hence p divides neither 3^{k+1} nor $2^{3^k} - 1$, hence p never divides $\text{ord}(\kappa_{3^k} + m)$ for $m \in \{0, 1, 2, 3\}$. By the order criterion, $m_p \geq 4$. The full primitivity conjecture C_k below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g. $p = 87211$, $f = 54$.*

Proposition 9 (the 3-power column). *[proved] For a prime r with $f(r) = 3^k$, the values $m \in \{0, 2, 3\}$ are impossible: $m_r = 0$ would require $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$, a power of 3, while $r \equiv 1 \pmod{3^k}$ is a prime distinct from 3; and $m = 2, 3$ share $m = 1$'s order class by Proposition 7. Any failure of C_k therefore gives $m_r \geq 4$ directly.*

Proposition 10 (norm tower). *[proved] $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$ (minimal polynomial $X^3 + X + \gamma_{k-1}$), so $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$. With lifting-the-exponent ($v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$ for old primes $r \neq 3$), full-order parts propagate upward, and conjecture C_k reduces to C_{k-1} plus primitivity at the new primes $r \mid \Phi_{3^k}(2)$.*

Conjecture 1 (C_k : γ -primitivity). γ_k is primitive in $L^\times = \mathbb{F}_{2^{3^k}}^\times$.

Proposition 11 (equivalence). *[proved] For primes r with $f(r) \mid 3^k$ a 3-power:*

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

The family formula is the 0/1/4 candidate of Section 4 restricted to the 3-power column.

3.2 Certification status

[**certified** $k \leq 6$] C_k holds for $k = 1, \dots, 6$: γ_k is primitive in $\mathbb{F}_{2^{3k}}^\times$, and $\text{ord}(\kappa_{3k} + 1) = 3^{k+1}(2^{3k} - 1)$ is verified exactly – product reconstruction, squarefreeness, primality by deterministic Miller–Rabin below 3.3×10^{24} and MR-64 probable-primality above, an independent sieve re-deriving the small factors, cross-validation against the independent term algebra for $k = 1, 2$ and against the calculator rows $m_{2593} = 1$, $m_{487} = 1$ (`experiments/cyclotomic_3k_family.py`, `experiments/excess/`).

[**consistent**] $k = 7, 8$: every known prime factor of $\Phi_{37}(2)$ and $\Phi_{38}(2)$ divides $\text{ord}(\gamma_k)$. Full certification is blocked only by the unfactored cofactors (FactorDB status composite-no-factor); no failure has been found.

[**open**] Whether ECM or GNFS can factor the $\Phi_{37}(2)$ and $\Phi_{38}(2)$ cofactors within a realistic budget, converting $k = 7, 8$ from CONSISTENT to CERTIFIED.

3.3 Certified excess rows produced

The following $m_r = 1$ rows are certified by the order criterion independently of the cofactor issue (analysis-level results; they are not new Rust carries):

f	primes r with $m_r = 1$
27	262657
81	71119, 97685839
243	16753783618801, 192971705688577, 3712990163251158343
729	80191, 97687, 379081, P42, P90
2187	39367, 7606246033, 263196614521, 529063556041
6561	209953, 1299079, 70063267397606709277393

(P42, P90 denote the certified/probable primes of 42 and 90 decimal digits dividing $\Phi_{36}(2)$.)

4 The 0/1/4 candidate

Conjecture 2 (the 0/1/4 rule). [**consistent**] (*not proved*)

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k, k \geq 1, \\ 1 & \text{otherwise (singleton odd prime-power } \mathcal{Q}(f(p))). \end{cases}$$

The rule matches all 950 calculator records with known $\mathcal{Q}$ -sets and every OEIS-known row covered by those $\mathcal{Q}$ -sets.

4.1 The component set alone is insufficient

[**proved**] $\mathcal{Q}(f(p))$ alone does not determine m_p :

$\mathcal{Q}$	$m = 4$	$m = 1$
$\{9\}$	$p = 19, f = 18$	$p = 73, f = 6$
$\{81\}$	$p = 163, f = 162$	$p = 2593, f = 18$
$\{243\}$	$p = 1459, f = 486$	$p = 487, f = 18$

[proved] The split for $\mathcal{Q} = \{9\}$ is explained by the order criterion: $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$ and $511 = 7 \cdot 73$, so $73 \mid \text{ord}(\kappa_9 + 1)$ but $19 \nmid \text{ord}(\kappa_9 + 1)$; adding $m = 4$ changes the order and introduces the factor 19.

[proved] The lower bound of the exception column is unconditional: $m_p \geq 4$ for all p with $f(p) = 2 \cdot 3^k$ (Proposition 8).

[conjectured] The matching upper bound $m_p \leq 4$ on that column; equivalent to C_k for the primes in the exception column.

4.2 The $m = 4$ upper bound: settled at every visible prime

The $m = 4$ translate $\kappa_{3^k} + 4$ involves the number $4 \in \mathbb{F}_{16}$, which does not lie in $\mathbb{F}_2(\zeta)$ (as $4 \nmid 2 \cdot 3^k$); the translate lives in the degree- $4 \cdot 3^k$ compositum $\mathbb{F}_{2^{4h}}$, $h = 3^k$. (June 2026 pass; probe `experiments/exception_column_m4.py`.)

Proposition 12 (the corrected norm). **[proved]** Let $\sigma = \text{Frob}^{2h}$ generate $\text{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$. Then $\sigma(4) = 5$, and with $\omega := \zeta^h = \kappa_2$,

$$N := N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

Proof. $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$ because $\gcd(4, 2 \cdot 3^k) = 2$, so $\sigma|_{\mathbb{F}_{16}}$ is the nontrivial element of $\text{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$, which swaps the two roots 4, 5 of the Artin–Schreier minimal polynomial $y^2 + y + \omega$ of 4 over $\mathbb{F}_4$. (Equivalently: $2h \equiv 2 \pmod{4}$ and the Frobenius orbit of 4 is $4 \rightarrow 6 \rightarrow 5 \rightarrow 7$, so $\sigma(4) = 4^{2^2} = 5$.) Then $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + (4 \otimes 5)$ and $4 \otimes 5 = 4^2 + 4 = \omega$ by the same Artin–Schreier relation. $\square$

Remark 3 (erratum). The 2026-06-10 draft of this note stated the norm as $(\kappa + 4)(\kappa + \text{Frob}^2(4)) = (\kappa + 4)(\kappa + 6)$. The conjugate $6 = 4^2$ is $\text{Frob}^1(4)$, one squaring short of $\text{Frob}^2(4) = 5$. The slip is visible without computation: $4 \otimes 6 = 14 \notin \mathbb{F}_4$, so $(\kappa + 4)(\kappa + 6) \notin \mathbb{F}_{2^{2h}}$ and cannot be a norm.

The corrected norm collapses the $m = 4$ test into the *same* sparse trinomial model $\mathbb{F}_2[x]/(x^{2h} + x^h + 1)$ as the rest of the 3-power analysis: no compositum arithmetic is needed.

Corollary 2 (in-field criterion). **[proved]** For a prime p with $f(p) = 2 \cdot 3^k$ and $v_p(2^{2h} - 1) = 1$, write $\overline{N} = N^{2^h}$ and $M_k := \overline{N}/N = N^{2^h-1} \in U$. Then

$$m_p = 4 \iff p \mid \text{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

Proof. $m_p \geq 4$ is Proposition 8, so $m_p = 4$ iff $\kappa + 4$ has no p -th root. By Propositions 1 and 2 with $E = 4h$, $f = 2h$ (and $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$ by lifting the exponent, since $p \nmid 2$), that holds iff $N^{(2^{2h}-1)/p} \neq 1$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders, $p \mid 2^h + 1$, and $(2^{2h} - 1)/p = (2^h - 1) \cdot ((2^h + 1)/p)$, the test equals $M_k^{(2^h+1)/p} \neq 1$. $\square$

Conjecture 3 (D_k : the column analogue of C_k). The prime-to-3 part of $\text{ord}(M_k)$ is full: $((2^h + 1)/3^{k+1}) \mid \text{ord}(M_k)$. Equivalently (given the per-prime squarefreeness audits): $m_p = 4$ for every prime p with $f(p) = 2 \cdot 3^k$.

[certified $k \leq 6$] D_k holds for $k = 2, \dots, 6$: every prime factor of the fully factored $\Phi_{2 \cdot 3^k}(2)$ divides $\text{ord}(M_k)$. Hence $m_p = 4$ exactly — not merely ≥ 4 — for every prime p with $f(p) \in \{18, 54, 162, 486, 1458\}$:

f	primes p with $m_p = 4$
18	19
54	87211
162	163, 135433, 272010961
486	1459, 139483, 10429407431911334611, 918125051602568899753
1458	227862073, 3110690934667, 216892513252489863991753, 1102099161075964924744009, P78

(P78 is the certified 78-digit prime of $\Phi_{2,3^6}(2)$; PRP-local here, factordb marks it proven.) The anchor rows $m_{19} = m_{163} = m_{1459} = 4$ (DiMuro / calculator) are reproduced, never assumed; the remaining eleven rows are new. Independent cross-checks: the term algebra of `experiments/ordinal_excess_probe.py` re-derives the $p = 19$ root pattern ($m = 0, \dots, 3$ root, $m = 4$ no root) and $m_{87211} = 4$; an explicit Artin–Schreier compositum model $\mathbb{F}[y]/(y^2 + y + \omega)$ verifies $\sigma(4) = 5$, the norm identity, and the full direct power test $(\kappa + 4)^{(2^{4h}-1)/p} \neq 1$ on the smallest prime of each level $k \leq 6$.

[**consistent**] $k = 7, 8$: every *known* prime factor of $\Phi_{2,3^7}(2)$ and $\Phi_{2,3^8}(2)$ — seven and five primes respectively; factordb CF entries verified locally by divisibility, primality, and order tests, the small ones re-derived by sieve, and the list cross-audited against the raw factordb API after an adversarial review caught a dropped 43-digit $k = 7$ prime — also has $m_p = 4$. An $m_p \geq 5$ example, if one exists, now hides strictly inside the unfactored cofactors.

Proposition 13 (twisted norm tower). [**proved**] *With $\eta = \zeta^3 = \kappa_{3^{k-1}}$,*

$$N_{\mathbb{F}_{2^{2 \cdot 3^k}}/\mathbb{F}_{2^{2 \cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2 \eta + 1,$$

which is neither $N_{k-1} = \eta^2 + \eta + \omega$, nor one of its Frobenius conjugates (their exponent pattern is $(2^{j+1}, 2^j)$), nor a scaling $N_{k-1}(\lambda\eta)$ with $\lambda \in \mu_3$.

Proof. The generator τ of the degree-3 step sends $\zeta \mapsto \zeta^c$ with $c = 2^{2 \cdot 3^{k-1}}$ a primitive cube root of unity modulo 3^{k+1} ; the cube roots are $1 + a3^k$, so $\tau(\zeta) = \omega^a \zeta$ ($a \neq 0$) while $\tau(\omega) = \omega$. Hence the norm is $\prod_{r^3=\eta} (r^2 + r + \omega)$ over the three cube roots r of η . Writing α, β for the two roots of $T^2 + T + \omega$ (so $\alpha + \beta = 1$, $\alpha\beta = \omega$), each factor is $(r + \alpha)(r + \beta)$, and the product collapses by the resolvent

$$\prod_{r^3=\eta} (r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2 \eta + 1,$$

since $\alpha^3 + \beta^3 = (\alpha + \beta)^3 + \alpha\beta(\alpha + \beta) = 1 + \omega = \omega^2$ and $(\alpha\beta)^3 = \omega^3 = 1$. Machine-checked at $k = 2, 3, 4$. $\square$

So, unlike the γ tower of Proposition 10, old-prime full-order parts do *not* propagate up the column, and D_k is genuinely per-level: the certification at each k stands on its own.

Remark 4 (the 3-part). [**consistent**] The 3-part of $\text{ord}(M_k)$ carries no column information but is strikingly regular in range: $v_3(\text{ord } M_k) = k + 1$ (full) for odd k and $k - 1$ (deficiency exactly 3^2) for even k , for $2 \leq k \leq 6$. No explanation is on record.

5 Boundedness

[**conjectured**] (Lenstra [5]) m_p is globally bounded. Lenstra gave unconditional lower-bound rules (singleton odd $\mathcal{Q}$ forces positive excess; $f(p) = 2 \cdot 3^k$ forces excess at least 4) but left absolute boundedness open.

[**proved**] If the 0/1/4 candidate holds, then $m_p \leq 4$ for all p ; proving Conjecture 2 would settle boundedness.

[**proved**] The 3-power column cannot produce $m \in \{2, 3\}$ (Proposition 9); any C_k failure jumps directly to $m_r \geq 4$.

[**proved**] (conditional on C_k) For every prime r with $f(r) = 3^j$, $j \leq k$: C_k implies $m_r = 1$. With C_k certified for $k \leq 6$, all primes in the 3-power column with $f(r) \leq 6561$ have $m_r = 1$ unconditionally.

[**certified** $k \leq 6$] (June 2026 pass) On the $2 \cdot 3^k$ exception column the matching upper bound is now certified at every visible prime: $m_p = 4$ exactly for every prime with $f(p) = 2 \cdot 3^k$, $k \leq 6$, and for every known prime at $k = 7, 8$ (Section 4.2). Boundedness on the exception column therefore holds unconditionally through $f = 1458$ and is equivalent to D_k (Conjecture 3) beyond.

[**open**] Boundedness outside the 3-power and $2 \cdot 3^k$ columns. The non-cyclotomic singleton chains (e.g. the 11-chain and the 23, 29, 47 components) have $m = 1$ rows in the calculator data and OEIS, but no order-criterion proof covers them the way C_k covers the 3-power column.

[**open**] Whether any prime p has $m_p \geq 5$. No counterexample is known; the question is not known to be decidable by any current structural argument outside the 3-adic columns.

6 Ranked next moves

1. *Dress rehearsal on m_{359} (highest priority).* The easy dependency rows are now locally certified: $m_{89} = 1$ at $E = 220$ and $m_{179} = 1$ at $E = 19,580$ (`experiments/ordinal_excess_probe.py`, `--deep` for the latter). The remaining independent rehearsal row on which an m_{719} certificate depends is $m_{359} = 1$, whose component field has degree $E = 3,504,820$. The value is already source-pinned by A380496; the research value is a local certificate and a calibrated cost model for the first unknown row. The 2026-06-16 pass found the precise obstruction: the cheap top-step norm lands in $\mathbb{F}_{2^{19580}}$, where 359 is invisible, while the correct norm is the transverse $\mathbb{F}_{2^{3504820}}/\mathbb{F}_{2^{179}}$ norm and is dense in the current term basis (Subsection 2.4). Certifying it remains a stretch, plausibly hours to a day with `gf2x`-style arithmetic, and is not reachable with the present pure-Python oracle. Confirm $m_{359} = 1$ first; the norm recurrence for $p = 719$ shifts if not. *Feasible only with a faster tower/Frobenius implementation; blocks the next move.*
2. *Certify m_{719} via the Kummer-tower Frobenius norm.* Implement the binary-splitting norm $N_{2k} = N_k \cdot \text{Frob}^{359k}(N_k)$ in the tower-sparse representation; measure the cost model on the calibrated path; if tower-aware Frobenius brings the ~ 44 modular compositions to feasible size, run to certify or falsify $m_{719} = 1$. Factor out the shared `relative_norm_over_frobenius_orbit` abstraction (Remark 2). *Blocked on move 1 for the cost model.*
3. *[Done, 2026-06-12.] $m_p = 4$ exactly on the $2 \cdot 3^k$ column.* Settled at every prime current factor tables reach (Section 4.2, `experiments/exception_column_m4.py`): the corrected norm $(\kappa+4)(\kappa+5) = \kappa^2 + \kappa + \omega$ collapses the test in-field; certified universally for $k \leq 6$, consistent at $k = 7, 8$. Successor moves: prove D_k (Conjecture 3) — the twisted norm tower (Proposition 13) shows why γ -style propagation is unavailable; factor the $\Phi_{2,37}(2)$ and $\Phi_{2,38}(2)$ cofactors (joins the ECM/GNFS batch of the next move); explain the 3-part parity (Remark 4).

4. *Convert C_7 , C_8 from CONSISTENT to CERTIFIED.* ECM/GNFS jobs on the cofactors of $\Phi_{37}(2)$ and $\Phi_{38}(2)$; full factorization also emits a batch of new A380496 rows. *Background job; uncertain outcome; blocks nothing.*
5. *Exact-order census on the non-cyclotomic singleton chains.* Compute $\text{ord}(\kappa_q + m)$ for small m in the explicit component fields for the 11-chain and the 23, 29, 47 components (degrees up to ~ 5060 ; trivial locally); look for an analogue of the $L^\times \times U$ splitting and for any forced-translate mechanism. This is the main structural gap for boundedness outside the 3-power columns. *Feasible now; no structural theory yet.*

7 Provenance and validation

7.1 Independent oracle

`experiments/ordinal_excess_probe.py` is a small local term-algebra oracle – not a replacement for CGSuite [8] or the C++ calculator [9], but a way to verify the first subtle cases without using the Rust production tower as its own oracle. It implements the impartial term algebra used by the calculators, a multiplicative-order test for small component fields, and a fixed-base exponentiation path (ported from the calculator’s strategy) for targeted root tests where full order factorization is unnecessary. Current probe output:

p	m	$\mathcal{Q}$	root?	
7	0	{3}	yes	
7	1	{3}	no	
19	1	{9}	yes	
19	4	{9}	no	
73	1	{9}	no	
89	1	{11}	no	
47	1	{23}	no	
179	1	{89}	no	(--deep)

The $p = 47$ line certifies $m_{47} = 1$ using only lower verified rows; since $f(47) = 23$ and $\mathcal{Q}(23) = \{23\}$, the carry is $\alpha_{47} = \kappa_{23} + 1 = \omega^{(\omega^7)} + 1$, the value now implemented in `tower.rs`. The $p = 179$ line is the first locally certified dependency row beyond the quick probe path; it still uses only the already-certified lower row $m_{89} = 1$.

7.2 The α_{47} promotion

Superseded 2026-06-13: this pass moved the boundary to α_{53} by promoting a single locally verified row; the finite excess table is now the full OEIS A380496 b-file (126 rows, odd primes $3 \leq p \leq 709$), so the refusal boundary sits at α_{719} and the boundary test is `boundary_returns_none_past_prime_709`. The original pass is recorded below for provenance.

The shipped state at the time: `src/scalar/big/ordinal/tower.rs` carries α_{47} with the landmark test `locally_verified_alpha_47_landmark` and moves the refusal boundary to α_{53} ; the documented boundaries in `nim.rs`, `mod.rs`, `src/games/nimber_game.rs`, `README.md`, and `docs/OPEN.md` were updated in step. The full gate ran clean: `cargo fmt --check`, `cargo test`, `cargo check/clippy` with and without the `python` feature (warnings denied), and the probe under `py_compile`.

7.3 Supporting probes

The 3^k -family certification chain is `experiments/cyclotomic_3k_family.py` (committed, maintained), joined 2026-06-12 by `experiments/exception_column_m4.py` (the $2 \cdot 3^k$ exception column, Section 4.2; committed, maintained, stdlib-only, ~ 2 minutes), with the research-run probes under `experiments/excess/` (machine-generated, rescued from ephemeral storage; triage before citing). The analysis in Sections 2–5 synthesizes the 2026-06-10 parallel research run plus the 2026-06-12 exception-column pass; each claim above carries its own tag, and no claim depends on an unverified solver.

Acknowledgements

The `ogdoad` library, the experiments, and this draft were developed with LLM assistance; the analysis consolidates a structured parallel research run (2026-06-10). Mathematical direction, framing, and responsibility for claims remain the author’s.

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